II PERIODIC TEST : 2025-26

Class - X

Subject - Artificial Intelligence (417)

Time : 2 hrs. M.M.: 50

General Instruction :

1. **Please read the instructions carefully.**
2. **This question paper consists of two sections: Section A and Section B**
3. **Section A has Objective type questions (MCQ, True/False and Fill in the blanks), whereas Section B contains Subjective type questions.**
4. **All questions of a particular section must be attempted in correct order.**
5. **Marks allotted are mentioned against each question/part.**

SECTION A : Objective Type Questions

Q.1- Multiple choice questions:- *[10]*

i. Which of the following is a limitation of rule-based AI model?
   a. They can update their rules automatically.
   b. They adapt to new data easily.
   c. They do not modify their rules after training.
   d. They do not require predefined rules.

ii. What role do neural networks play in Deep Learning?
   a. They create static algorithms.
   b. They process large amount of data and recognize patterns.
   c. They identify anomalies in simple datasets.
   d. They only function for rule-based systems.

iii. What is the purpose of online evaluation?
   a. To test a model before deployment.
   b. To track model performance in real-time.
   c. To delete unnecessary data from the dataset.
   d. To track model on real-world data.

iv. Which factor can cause AI models to make incorrect predictions?
   a. High-quality training data.
   b. Proper error analysis.
   c. Poor-quality data.
   d. 100% accuracy.

v. What is the primary role of data science in decision-making?
   a. To predict future outcomes.
   b. To summarise past data.
   c. To make informed, evidence-based decisions.
   d. To analyse any visual data.

vi. What is one major limitation of No-Code AI tootls?
   a. They require extensive coding knowledge.
   b. They lack flexibility for customization.
   c. They are only used in gaming applications.
   d. They are extremely expensive.

vii. Which of the following is an example of a No-Code AI tools?
   a. Python
   b. JavaScript
   c. Orange Data Mining
   d. TensorFlow

viii. Which of the following statements is true about Jupyter Notebook?
   a. Anaconda is web-based tool for installing Jupyter Notebook.
   b. Anaconda is a command-line interface tool for installing Jupyter Notebook.
   c. Anaconda can only be used from a web browser to install Jupyter Notebook.
   d. Anaconda is used to write Jupyter Notebooks, not install them.

Contd.....2-

ix. How do you access the last character of a string using negative indexing?
   a. String[-0] b. String[-1]
   c. String[-2] d. String[-3]

x. What is the main advantage of using NumPy arrays over Python lists for numerical computations?
   a. They can store mixed data types.
   b. They support direct arithmetic operations.
   c. They consume more money.
   d. They are slower for computations.

Q.2- ***Fill in the blanks:-*** ***[10 × 1 = 10]***

i. AI systems rely on large amounts of _____ to make accurate decisions.
ii. The process where a neural network adjusts its weights based on errors is called _____ .
iii. In Deep Learning, _____ are used to process information and recognize patterns.
iv. The proportion of correct predictions made by an AI model is called _____ .
v. _____ tuning helps improve a model's accuracy by optimizing its setting like learning rate and number of layers.
vi. _____ occurs when model memorises patterns from training data but fails to generalize to new data.
vii. Machine learning is a subset of artificial intelligence that focuses on developing algorithms to enable machines to make _____ or decision based data.
viii. Data science involves collecting, analyzing, and using data to solve problems or make _____ decisions.
ix. The _____ attribute of sort( ) method is used to reverse the order of the list.
x. NumPy arrays are _____ because they can only store one type of data.

Q.3- ***State True False:-*** ***[10 × 1= 10]***

i. A testing dataset is used to train the AI models.
ii. The output layer of a neural network processes data before presenting it to the user.
iii. Error analysis is not necessary for improving AI models.
iv. AI can only be used in computers and not in mobile devices.
v. AI helps machines make decisions based on data.
vi. Machine learning is a part of data science that helps analyse past data to inform future decisions.
vii. Teachable Machine allows users to create machine learning models without coding.
viii. The mean and median are always same in a normal distribution.
ix. You can access tuple entry using indexing.
x. Lists cannot store different types of elements.

SECTION B : Subjective Type Questions

Short Answer Questions:-

Q.4- Why is backward propagation important in a neural network? [2]
Q.5- What is the significance of errors in AI model? [2]
Q.6- What are two advantages using No-Code AI? [2]
Q.7- What is the purpose of the np.linspace( ) function? [2]

Long Answer Questions:-

Q.8- What factors contribute to errors in AI models, and how can they be minimized? [3]
Q.9- Describe the types of data distributions and provide examples for each. [3]

Python Programs:-

Q.10- Create a DataFrame that contains information about three individuals, including their names, ages, and cities of residence. The DataFrame should have the following columns :
   Name : Rahul, Amit, and Sandeep
   Age : 38, 40, and 42
   City : Sitapur, Hardoi and Kanpur [3]

Q.11- Write a program to create a bar graph showing the number of fruit sold. The data is as follows :
   Mangos : 10
   Apples : 5
   Oranges : 11 [3]

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